

Finding the Error: Correspondence and Scale Factor

Directions

- Each problem shows a piece of work that reaches a wrong answer. The arithmetic in it is always correct — the mistake is in how the proportion or the scale factor was set up.
- First redo the problem correctly and write the corrected value, with its unit, on the answer line.
- Then name the error in a sentence: say what was matched with what, and what should have been matched instead.
- A similarity statement is read position by position: in $\triangle ABC \sim \triangle DEF$, A matches D , B matches E , C matches F .

1. $\triangle ABC \sim \triangle DEF$ with $AB = 6$ cm, $BC = 8$ cm, and $DE = 9$ cm. A student wrote

$$\frac{6}{9} = \frac{9}{EF} \quad \text{so} \quad EF = 13.5 \text{ cm.}$$

- (a) Find the correct length of $\overline{EF}$, in centimetres.

Answer: _____ cm

- (b) Name the error in the student's proportion.

- 2.** $\triangle PQR \sim \triangle STU$ with $PQ = 10$ in, $QR = 15$ in, and $ST = 4$ in. A student wrote

$$\frac{10}{4} = \frac{TU}{15} \quad \text{so} \quad TU = 37.5 \text{ in.}$$

- (a) Find the correct length of $\overline{TU}$, in inches.

Answer: _____ in

- (b) Name the error in the student's proportion.

- 3.** $\triangle ABC \sim \triangle XYZ$ with $AB = 12$ m, $BC = 20$ m, and $XY = 18$ m. A student wrote

$$\text{scale factor from } \triangle ABC \text{ to } \triangle XYZ = \frac{12}{18}, \quad YZ = 20 \cdot \frac{12}{18} = 13.33 \text{ m.}$$

- (a) Find the correct scale factor from $\triangle ABC$ to $\triangle XYZ$.

Answer: _____

- (b) Find the correct length of $\overline{YZ}$, in metres.

Answer: _____ m

- (c) Explain what went wrong with the student's scale factor.

4. $\triangle JKL \sim \triangle MNP$ with $m\angle J = 40^\circ$ and $m\angle K = 75^\circ$. A student concluded that $m\angle P = 75^\circ$.

(a) Find the correct value of $m\angle P$, in degrees.

Answer: _____

(b) Name the error in the student's reasoning.

5. A poster shop reduces a triangular decal. The original decal $\triangle ABC$ has $AB = 30$ cm and $BC = 42$ cm; the reduced copy $\triangle DEF$ has $DE = 7.5$ cm. A student wrote

$$EF = 42 \cdot \frac{30}{7.5} = 168 \text{ cm.}$$

(a) Find the correct length of $\overline{EF}$, in centimetres.

Answer: _____ cm

(b) Explain which way the scale factor has to be applied here, and how the student's answer shows the direction was reversed.

6. $\triangle ABC \sim \triangle DEF$ with $AB = (x + 4)$ cm, $DE = 12$ cm, $BC = 10$ cm, and $EF = 15$ cm. A student wrote

$$\frac{x + 4}{12} = \frac{15}{10} \quad \text{so} \quad x = 14.$$

(a) Solve correctly for x .

Answer: _____

(b) Name the error in the student's proportion.